Topological evaluation of Causal and Temporal reasoning in Large Vision-Language Models

Sviluppo di un benchmark basato su Task Graph per l'analisi di video procedurali

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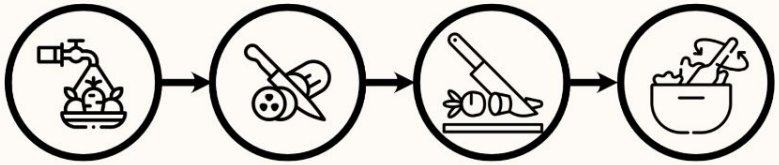
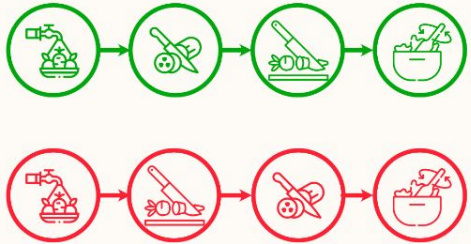
Shuffle & Reorder

I task reali non sono lineari

- Assunzione attuale: un'unica sequenza corretta.
- Incompatibile con le procedure reali (es. ricette in cucina).
- Penalizza varianti operative logicamente valide.

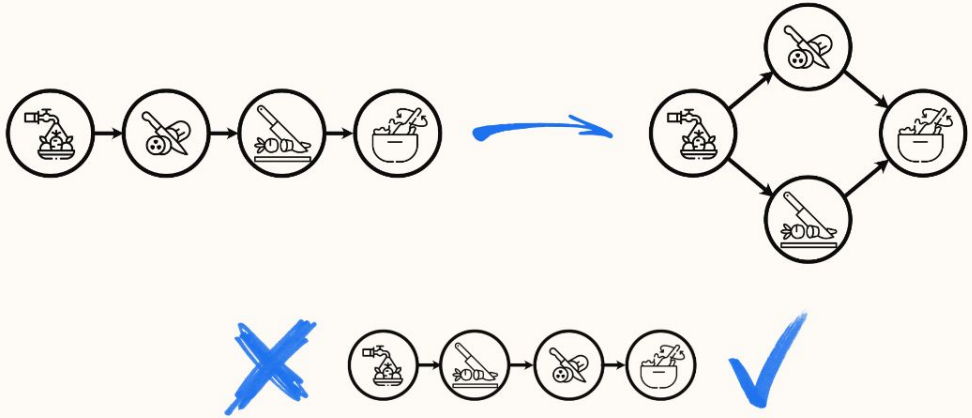




La soluzione?

Coerenza topologica

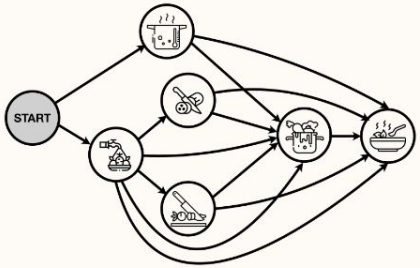
- Rappresentazione tramite Task Graphs (DAG)
- Separazione netta tra ordine di esecuzione e l'ordine di dipendenza



Le metriche

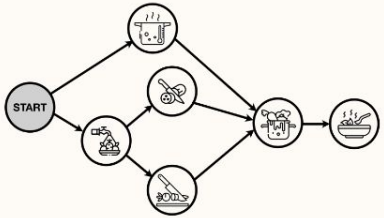
Closure Accuracy

$$\mathrm{ClosureAcc}(\hat{y}, G^*) = \frac{\sum_{(u,v)\in E(G^*)} \mathbb{I}[pos_{\hat{y}}(u) < pos_{\hat{y}}(v)]}{|E(G^*)|}$$



Reduction Accuracy

$$\mathrm{ReductionAcc}(\hat{y}, G_{red}) = \frac{\sum_{(u,v)\in E(G_{red})} \mathbb{I}[pos_{\hat{y}}(u) < pos_{\hat{y}}(v)]}{|E(G_{red})|}$$



Esperimenti: analisi e risultati

Il setup sperimentale

Dataset: Captain Cook4D e COMKITCHENS.

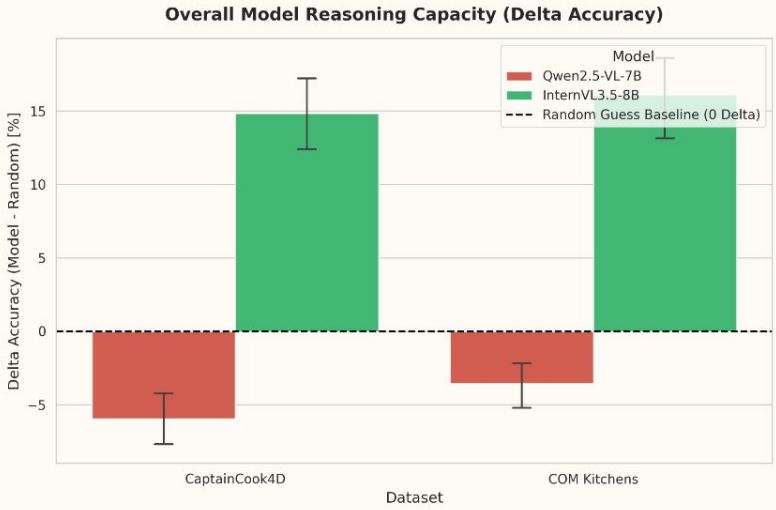
Modelli: InternVL3.5-8B e Qwen2.5-VL-7B

Variabili: Clip (3-5-7-9 clip) e Frame (2-4-6-8 frame).

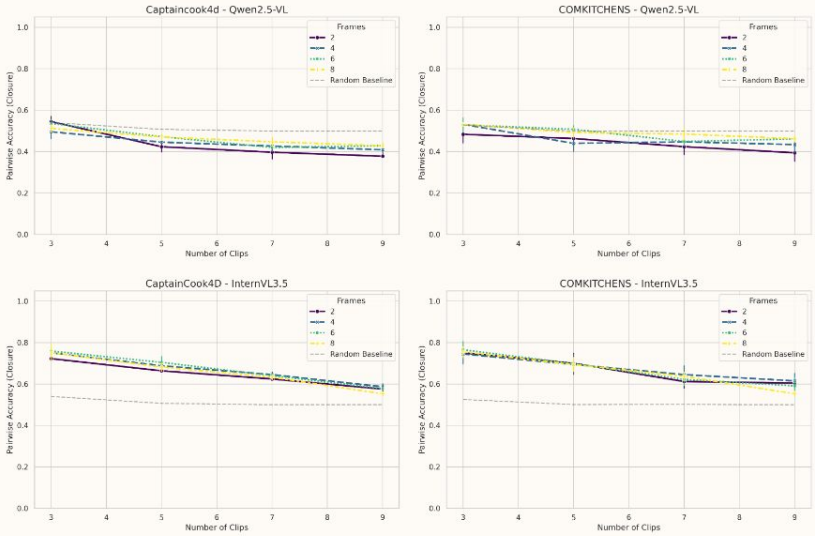
Captaincook4d

COMKITCHENS

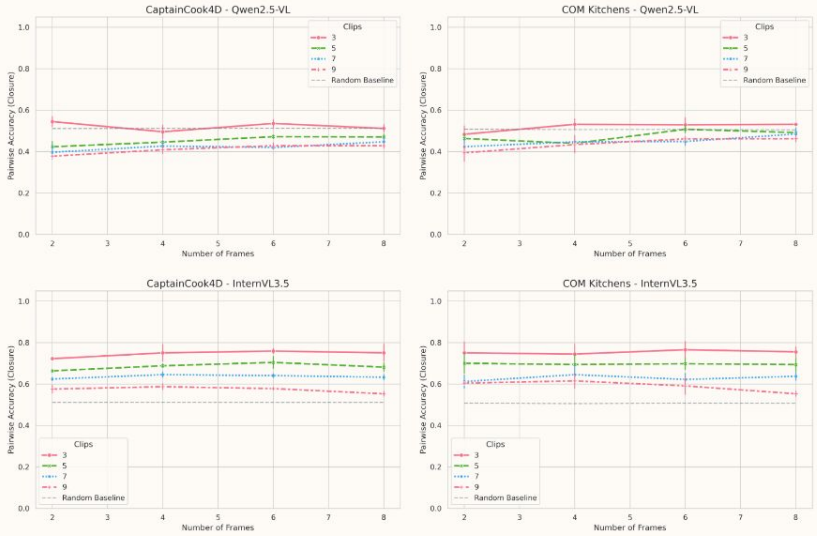
Il divario tra i modelli



Il paradosso della densità visiva



Variazione per clip



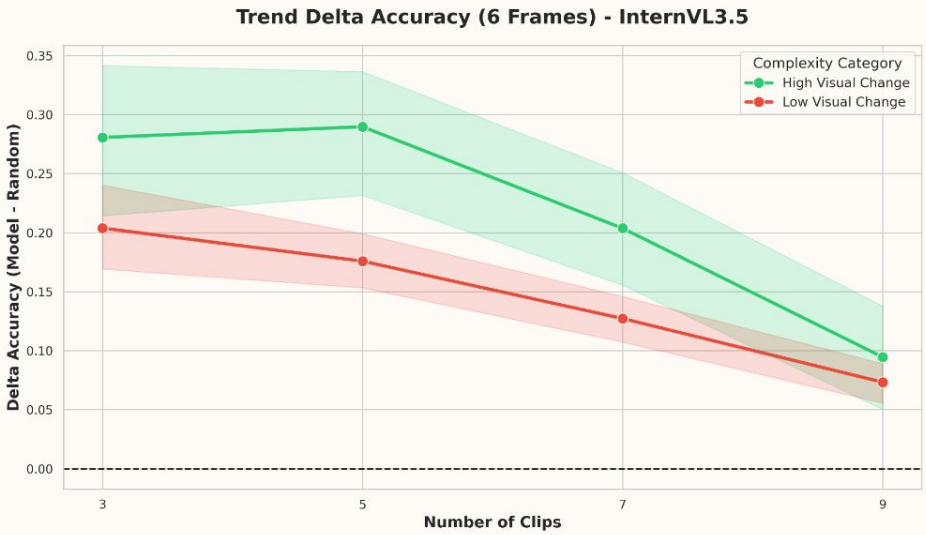
Variazione per frame

Analisi qualitativa

High Visual Change: cambiamenti di stato chiari e irreversibili e progressioni identificabili.

Low Visual Change: variazioni visive minime e/o azioni ambigue.

I modelli sfruttano indizi visivi e *visual bias*



High Visual Change

Low Visual Change

Conclusioni: takeaways e sviluppi futuri

Grazie

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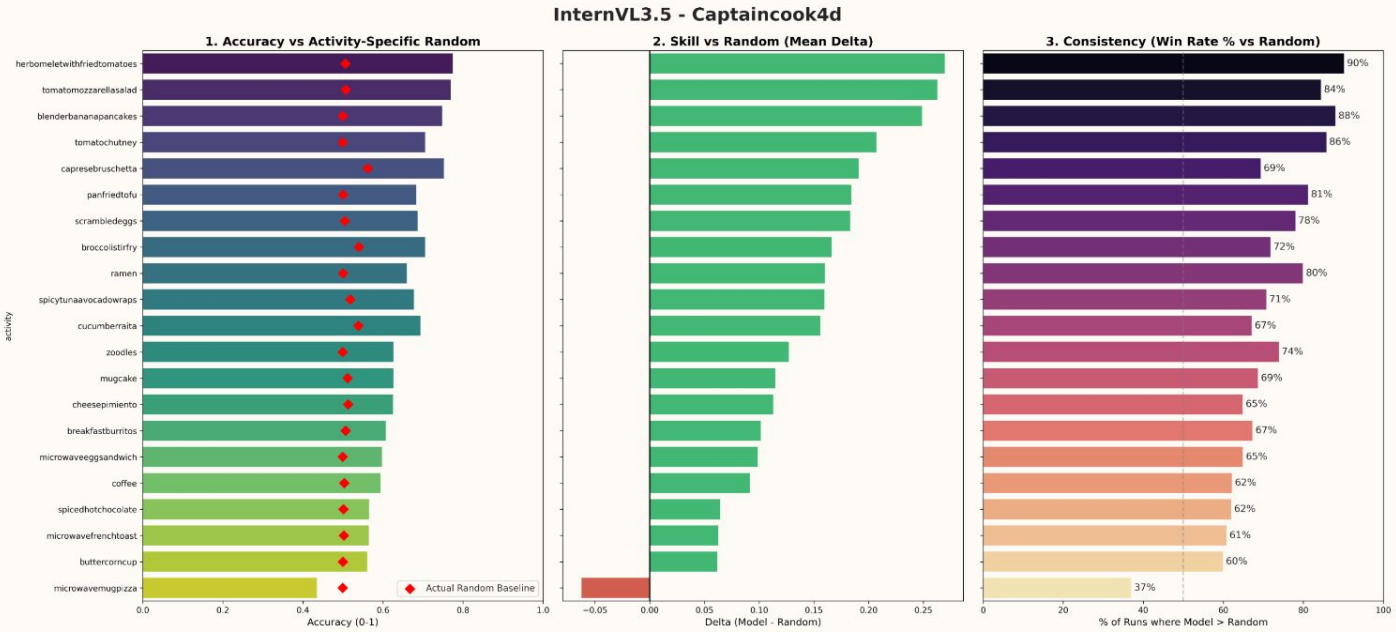
Tabelle dei risultati

Frames	Clips	Model	CaptainCook4D		COMKITCHENS	
			Closure	Reduction	Closure	Reduction
2	3	Qwen2.5-VL-7B-Instruct	54.5 ± 2.4	54.7 ± 2.6	48.3 ± 4.1	49.8 ± 4.9
		InternVL3.5-8B	72.2 ± 0.5	70.3 ± 0.6	75.0 ± 5.2	73.4 ± 6.4
		Random	53.9	53.9	53.0	53.1
2	5	Qwen2.5-VL-7B-Instruct	42.3 ± 2.6	42.3 ± 2.4	46.3 ± 2.2	45.5 ± 1.5
		InternVL3.5-8B	66.4 ± 0.6	62.1 ± 1.3	70.0 ± 4.8	65.6 ± 4.0
		Random	50.7	50.6	50.1	50.1
2	7	Qwen2.5-VL-7B-Instruct	39.7 ± 3.2	39.0 ± 3.3	42.4 ± 3.8	41.8 ± 3.2
		InternVL3.5-8B	62.4 ± 0.8	58.3 ± 1.0	61.2 ± 3.2	58.0 ± 1.8
		Random	49.9	49.9	50.0	50.0
2	9	Qwen2.5-VL-7B-Instruct	37.8 ± 0.7	38.2 ± 0.5	39.4 ± 4.1	39.8 ± 3.9
		InternVL3.5-8B	57.6 ± 1.9	53.7 ± 1.2	60.4 ± 0.7	55.8 ± 0.7
		Random	50.0	49.9	49.9	50.0
4	3	Qwen2.5-VL-7B-Instruct	49.5 ± 3.2	49.9 ± 3.0	53.2 ± 2.7	54.2 ± 2.2
		InternVL3.5-8B	75.0 ± 4.0	73.6 ± 4.1	74.5 ± 4.7	72.6 ± 4.6
		Random	54.0	53.9	52.3	52.4
4	5	Qwen2.5-VL-7B-Instruct	44.5 ± 0.4	44.3 ± 0.1	43.9 ± 3.8	42.9 ± 2.4
		InternVL3.5-8B	68.8 ± 1.0	63.7 ± 0.7	69.4 ± 4.9	66.2 ± 4.5
		Random	50.7	50.7	49.9	49.9
4	7	Qwen2.5-VL-7B-Instruct	42.7 ± 2.8	42.2 ± 3.1	44.7 ± 2.2	44.2 ± 1.9
		InternVL3.5-8B	64.5 ± 1.4	60.3 ± 1.4	64.5 ± 4.2	60.1 ± 3.6
		Random	49.9	50.0	50.1	50.1
4	9	Qwen2.5-VL-7B-Instruct	40.9 ± 2.0	41.6 ± 1.4	43.4 ± 4.1	44.0 ± 4.2
		InternVL3.5-8B	58.7 ± 1.4	53.9 ± 2.7	61.5 ± 3.6	57.1 ± 3.6
		Random	50.0	50.0	50.0	50.0

Frames	Clips	Model	CaptainCook4D		COMKITCHENS	
			Closure	Reduction	Closure	Reduction
6	3	Qwen2.5-VL-7B-Instruct	53.6 ± 1.8	53.4 ± 1.5	52.9 ± 3.5	54.0 ± 1.7
		InternVL3.5-8B	75.9 ± 1.3	73.4 ± 0.5	76.5 ± 4.0	74.9 ± 3.5
		Random	54.1	54.1	52.7	52.8
6	5	Qwen2.5-VL-7B-Instruct	47.2 ± 0.8	46.7 ± 1.5	50.7 ± 1.7	48.9 ± 1.2
		InternVL3.5-8B	70.4 ± 2.9	66.1 ± 3.2	69.8 ± 2.9	66.0 ± 3.5
		Random	50.7	50.6	50.1	50.1
6	7	Qwen2.5-VL-7B-Instruct	42.0 ± 0.8	42.2 ± 0.7	44.9 ± 1.9	44.9 ± 1.6
		InternVL3.5-8B	64.1 ± 0.9	59.4 ± 1.8	62.2 ± 2.9	58.7 ± 1.0
		Random	50.0	49.9	50.1	50.0
6	9	Qwen2.5-VL-7B-Instruct	42.8 ± 1.4	44.2 ± 1.2	46.1 ± 1.3	46.2 ± 1.8
		InternVL3.5-8B	57.8 ± 0.6	53.2 ± 1.1	59.1 ± 4.1	54.2 ± 3.7
		Random	50.0	49.9	49.9	50.0
8	3	Qwen2.5-VL-7B-Instruct	51.2 ± 1.8	50.5 ± 2.0	53.2 ± 0.6	54.0 ± 2.3
		InternVL3.5-8B	75.1 ± 4.1	73.0 ± 4.7	75.5 ± 2.4	73.1 ± 3.0
		Random	53.9	53.9	52.6	52.6
8	5	Qwen2.5-VL-7B-Instruct	47.1 ± 1.4	47.2 ± 1.5	49.2 ± 2.3	47.9 ± 2.8
		InternVL3.5-8B	68.2 ± 2.0	63.5 ± 1.6	69.4 ± 3.3	65.2 ± 3.1
		Random	50.7	50.7	49.8	50.0
8	7	Qwen2.5-VL-7B-Instruct	44.7 ± 2.3	44.9 ± 1.4	48.5 ± 2.8	48.6 ± 1.5
		InternVL3.5-8B	63.3 ± 1.2	59.1 ± 0.8	63.7 ± 2.0	60.4 ± 1.6
		Random	50.0	50.0	50.0	50.1
8	9	Qwen2.5-VL-7B-Instruct	42.8 ± 1.2	43.6 ± 2.1	46.2 ± 1.5	46.6 ± 1.1
		InternVL3.5-8B	55.3 ± 1.1	50.7 ± 2.0	55.3 ± 1.2	52.0 ± 1.7
		Random	50.0	49.9	50.0	49.9

Leaderboard ricette



Dataset considerati

- CaptainCook4D
- Comkitchens
- **proMQA Assembly** (Assembly101)
  Attività di costruzione troppo *fine grained*
- **STAR** (Charades)
  Maggior parte attività troppo semplici/atomiche
- **Graph2Vid** (CrossTask)
  Non esiste più il file con annotazioni

Tabella A.1: Confronto tra i due dataset selezionati per la fase sperimentale.

Caratteristica	**CaptainCook4D**	**COMKITCHENS**
Video totali (usati)	384 (173)	145 (67)
Ricette (usate)	24 (21)	139 (139)
Durata media video	14.7 min	16.6 min
N. medio clip per video	∼13	∼20
Durata media clip	1 min	46.7 s
Prospettiva	Egocentrica	Overhead
Struttura Task Graph	Ricavata da sequenze	Visual Action Graph (VAG)

Prompt utilizzato

```
You are given [N] shuffled segments of a video demonstrating the activity: '[ACTIVITY_NAME
    ]'.

Each segment has a unique ID ([ID_1, ID_2, ..., ID_N]).

Segment ID [ID_1]:
<image><image>...<image>

Segment ID [ID_2]:
<image><image>...<image>
...

Your Goal: Reorder these segments into the correct chronological sequence for '[
    ACTIVITY_NAME]'.

Instructions:
ANALYZE: Briefly describe what is happening in each Segment ID.
REASON: Explain the logical order based on cooking procedure.
SOLVE: Output the final sequence.

STRICT OUTPUT FORMAT:
Analysis: [Brief descriptions]
Reasoning: [Step-by-step logic]
Final Sequence: [ID_X]->[ID_Y]->... (Use actual IDs)
```